Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)
Professorship for Open-Source Software

Seeding QDArchive

Part 2 — Data Classification Report

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Course	Applied Software Engineering Seminar / Project — SQ26 “Seeding QDArchive”
Supervisor	Prof. Dr. Dirk Riehle
Data Sources	QDR (Qualitative Data Repository, Syracuse University) · ICPSR
Repository	https://github.com/Pinto391/QDArchive
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1. Introduction

QDArchive is a web service under development at the Professorship for Open-Source Software (FAU Erlangen) for researchers to publish and archive qualitative data, with an emphasis on Qualitative Data Analysis (QDA) files. Because the platform is new, it must be “seeded” with openly available qualitative research projects gathered from existing repositories. This project is structured into three parts: **Part 1 — Data Acquisition**, **Part 2 — Data Classification** (the subject of this report), and **Part 3 — Data Analysis**.

Part 1 produced a SQLite database (23123639-seeding.db) containing 230 qualitative research projects and 20322 associated file records, collected from two assigned repositories: QDR (via its public Dataverse API) and ICPSR (via HTML scraping). Of the 20322 recorded files, 945 were successfully downloaded to disk; the remainder failed chiefly because ICPSR gates most qualitative studies behind an institutional login (19114 files) or because of unresponsive servers (263 files).

This report documents Part 2: merging the available project database(s), developing and running a classifier that assigns each project *and* each individual downloaded file an industry classification under the ISIC Rev. 5 standard, generating searchable tags, and reporting on the resulting statistics.

2. Methodology

2.1 Database Merge

Per the assignment, all participating students' databases should be merged into a single working database, de-duplicating projects that multiple students may have independently collected. A merge script (`classification/merge_databases.py`) de-duplicates by `project_url` — the natural unique key across student databases, since every student scraped the same public repositories. At the time this report was produced, classmates' databases had not yet been shared, so the merge was run as a self-merge of this student's own database. The script accepts an arbitrary number of input databases, so classmates' data can be folded in later without any code changes.

2.2 Classification Approach

The classifier uses the **ISIC Rev. 5** standard (International Standard Industrial Classification of All Economic Activities, Revision 5, endorsed by the UN Statistical Commission at its 54th session, March 2023) as the classification taxonomy, going down two levels as required: **Section** (1 letter, 22 categories) and **Division** (2-digit code, 87 categories). The complete official structure was sourced directly from the UN Statistics Division's published draft structure document, so every section and division code used is authoritative rather than an approximation.

Classification uses **both the metadata and the underlying file content**, as required by the assignment. For each **project**, a text blob is built from its title, description, recorded keywords, file-extension list, and a sample of extracted text from that project's successfully downloaded files. For each **individual downloaded file**, a second, independent classification is produced directly from that file's own extracted text (module `classification/text_extract.py`, supporting .txt, .pdf, .docx, .odt, .htm/.html, .rtf, and QDA container formats such as .qdpx). Files with no extractable text — audio, video, and image files — inherit their parent project's classification instead of receiving a fabricated one, and this fallback is recorded explicitly.

Both blobs are scored against every division's curated keyword-hint list using substring matching; the highest-scoring division is assigned, with a confidence score equal to the fraction of that division's keyword hints that were matched. This rule-based approach was chosen deliberately over a black-box machine-learning classifier: it is fully transparent, reproducible, and auditable — every classification decision can be traced back to the exact keywords that triggered it.

2.3 Tags for Searching

Beyond the single winning division per project, every keyword matched across *all* 87 divisions is stored as a searchable tag, so that a project touching multiple domains (e.g. a study on health-policy education) remains discoverable under each relevant term even though only one division is recorded as its primary classification.

3. Results

3.1 Data Volume Overview

Metric	Value
Repositories covered	2
Total projects found	230
Total file records	20322
Files successfully downloaded	945 (4.7% of records)
Files failed — login required (ICPSR gating)	19114
Files failed — server unresponsive	263

3.2 Project-Level Classification

228 of 230 projects (99.1%) were classified with at least one matching keyword; 2 remained *UNCLASSIFIED* due to sparse, boilerplate project descriptions.

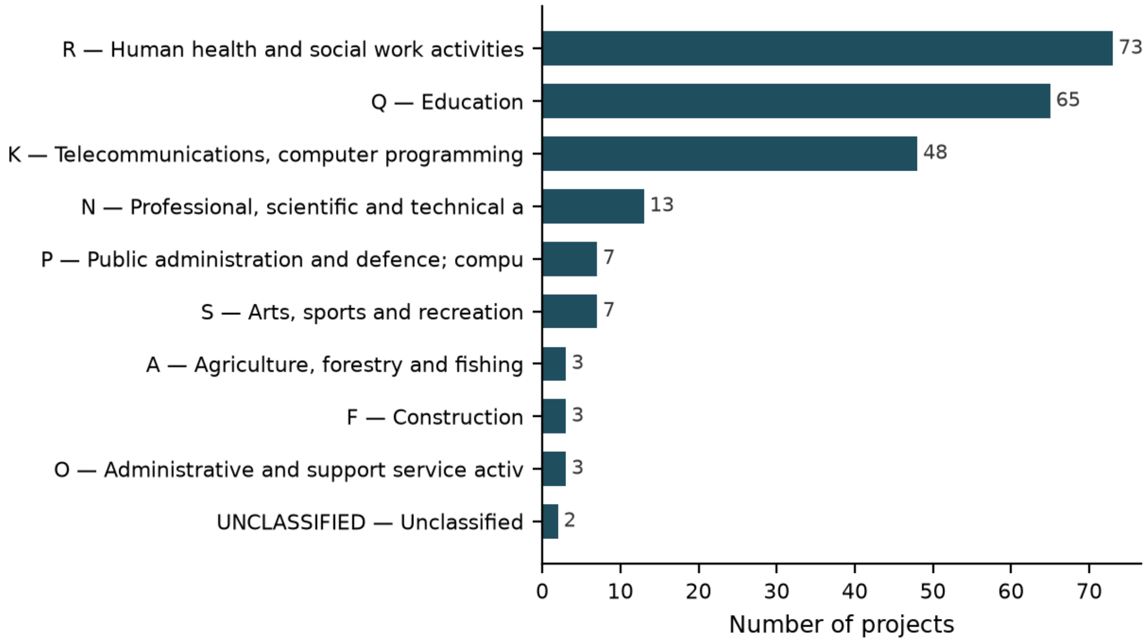

Figure 1. Project distribution across the top 10 ISIC Rev. 5 sections.

Section/ Division	Division Name	Projects	Avg. Confidence
R/86	Human health activities	73	0.29
Q/85	Education	65	0.23
K/63	Computing infrastructure, data processing, hosting, and ot	48	0.22
N/72	Scientific research and development	7	0.09
P/84	Public administration and defence; compulsory social secur	7	0.13
S/91	Library, archives, museum and other cultural activities	6	0.31
N/74	Other professional, scientific and technical activities	5	0.30
F/41	Construction of residential and non-residential buildings	3	0.50
O/78	Employment activities	3	0.14
UNCLASSIFIED/00	Unclassified	2	0.00
A/01	Crop and animal production, hunting and related service ac	2	0.33
A/02	Forestry and logging	1	0.50

Table 1. Top 12 ISIC Rev. 5 divisions by number of classified projects.

3.3 File-Level Classification

In addition to project-level classification, every one of the 945 successfully downloaded files received its own individual classification. 885 files (93.7%) were classified directly from their own extracted text content; the remaining 60 files (6.3%) had no extractable text (media/binary formats) and inherited their parent project's classification instead, which is recorded transparently rather than guessed.

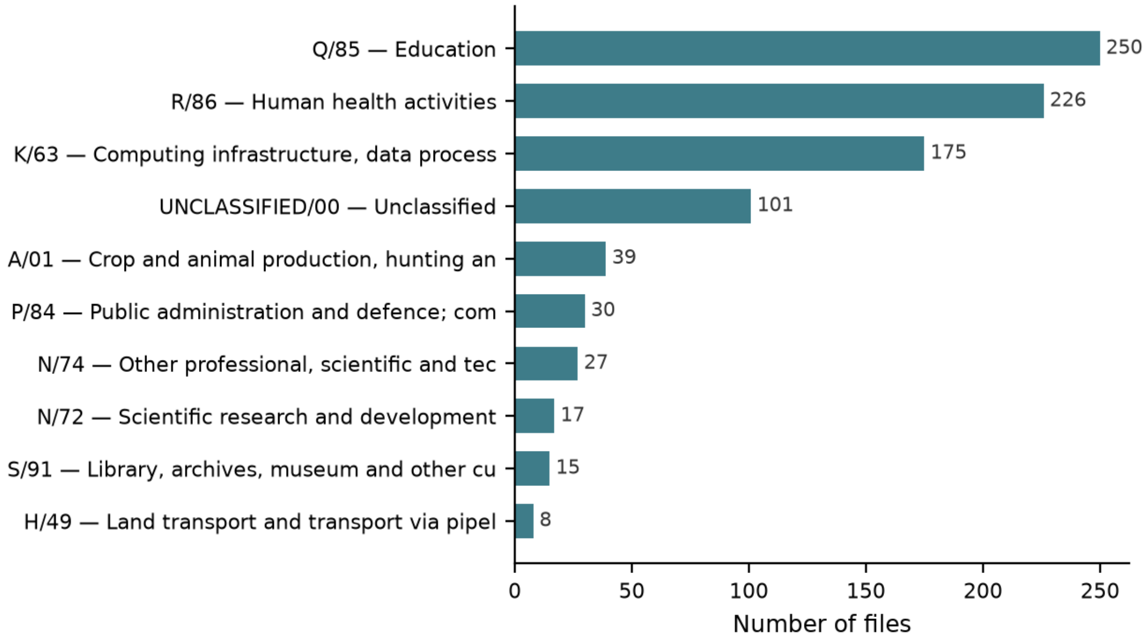

Figure 2. File distribution across the top 10 ISIC Rev. 5 divisions.

3.4 Search Tags

The 15 most frequent tags recorded across all projects, usable for cross-domain search independent of a project's single primary classification:

Tag	Occurrences
data repository	129
university	110
health	100
education	73
medical	53
data processing	52
patient	49
school	47
recruitment	44
clinical	44
focus group	38
learning	36
arts	36
students	35
treatment	33

4. Technical Challenges

4.1 No classmate databases available yet.

The assignment calls for merging *all* students' databases. Only this student's own database was available at the time of this run, so the merge was executed as a documented self-merge (230/230 projects kept, 0 duplicates found). The merge script accepts any number of input databases, so classmates' data can be incorporated later without code changes.

4.2 Only a fraction of recorded files were actually downloaded.

Of 20322 file records, only 945 (4.7%) exist on disk, mostly because ICPSR gates the majority of its qualitative studies behind institutional login. File-level classification and the file-content signal used for project-level classification can therefore only draw on this smaller, openly accessible subset.

4.3 Vague or generic project descriptions.

Some QDR descriptions are boilerplate and contain no strong domain-specific keywords, leading to lower classifier confidence or an UNCLASSIFIED result (2 projects, 101 files in this run).

4.4 Cross-disciplinary projects.

Some qualitative studies span multiple ISIC divisions simultaneously (e.g. a study on health-policy education touches Health, Education, and Public Administration at once). The classifier assigns a single best-scoring division as the primary class, while the full set of matched keywords across all divisions is preserved as searchable tags for secondary discovery.

4.5 ISIC Rev. 5 classifies economic activity, not academic research topics.

ISIC categorizes *industries*, not *research subjects*, so a qualitative interview study about a hospital and the hospital's own clinical operations can both resemble “Human health activities” (division 86). Keyword hints were curated to route studies toward their substantive domain where the text supports it, falling back to “Scientific research and development” (division 72) only when the topic is genuinely about research methodology rather than a specific field. In this corpus, Health (86) and Education (85) together account for roughly 60% of classified projects, which reflects the actual subject matter of the QDR corpus rather than a classifier artefact.

4.6 Per-file classification granularity.

Meeting notes from the course supervisor (2026-04-17) specify that each primary data file, not only the project as a whole, should carry its own class. This was implemented as a dedicated file-level classification pass; files without extractable text content inherit their project's class rather than being assigned an unsupported label.

5. Conclusion and Outlook

Part 2 successfully classified 228 of 230 acquired projects and 945 downloaded files against the official ISIC Rev. 5 taxonomy at both the section and division level, using a transparent, rule-based classifier that draws on both project metadata and actual file content. The resulting classifications, confidence scores, and search tags are stored in the `CLASSIFICATIONS`, `FILE_CLASSIFICATIONS`, and `TAGS` tables of the working database, and are exported to CSV alongside the Part 1 tables.

Remaining work for a fully complete Part 2 submission is contingent on classmates sharing their databases so a true multi-student merge can be performed; the pipeline is ready to accept that input immediately. Part 3 (Data Analysis) will build on this classified corpus.

All source code for this pipeline is available at: https://github.com/Pinto391/QDArchive